

Experiment – 04

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Subject: Advanced Database
Management System

Question 1:

A phone call is considered an international call when the person calling is in a different country than the person receiving the call.

What percentage of phone calls are international? Round the result to 1 decimal.

Assumption:

- The **caller_id** in **phone_info** table refers to both the caller and receiver.

phone_calls Table:

Column Name	Type
caller_id	integer
receiver_id	integer
call_time	timestamp

phone_calls Example Input:

caller_id	receiver_id	call_time
1	2	2022-07-04 10:13:49
1	5	2022-08-21 23:54:56
5	1	2022-05-13 17:24:06
5	6	2022-03-18 12:11:49

phone_info Table:

Column Name	Type
caller_id	integer
country_id	integer
network	integer
phone_number	string

phone_info Example Input:

caller_id	country_id	network	phone_number
1	US	Verizon	+1-212-897-1964
2	US	Verizon	+1-703-346-9529
3	US	Verizon	+1-650-828-4774
4	US	Verizon	+1-415-224-6663
5	IN	Vodafone	+91 7503-907302
6	IN	Vodafone	+91 2287-664895

Example Output:

international_calls_pct

50.0

Solution:

```
SELECT
  ROUND(
    COUNT(
      CASE
        WHEN P1.COUNTRY_ID <> P2.COUNTRY_ID THEN 1
      END
    ) * 100.0 / COUNT(*),
    1
  ) AS INTERNATIONAL_CALLS_PCT
FROM PHONE_CALLS AS C
JOIN PHONE_INFO AS P1
  ON C.CALLER_ID = P1.CALLER_ID
```

```
JOIN PHONE_INFO AS P2  
ON C.RECEIVER_ID = P2.CALLER_ID;
```